MILITARY SPECIFICATION

DETECTOR-DEWAR DT-617/UA

This amendment forms a part of Military Specification MIL-D-49419(ER), dated 18 January 1982.

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3.6.1 Delete 3.6.1 in its entirety and substitute the following:

3.6.1 Detector-Dewar defects. Unless otherwise specified herein, the detector-Dewar shall be operated at 80 (+5, -20) K. The detector-Dewar shall contain no more than 12 defective elements as defined in 6.3. Of the 12 defective elements, no more than 5 may be hard defects. The central 18 elements of the array shall contain no hard defects and no more than 2 soft defects. The central 36 elements shall contain no more than 1 hard defect and 4 soft defects. Hard defects shall be separated from one another by 4 elements and shall be electrically strapped to an adjacent element with common ground. Soft defects located within the central 36 elements shall be separated from any other defect by 4 nondefective channels and shall not be strapped unless the strapping is from an adjacent hard defect at the end of a group of 5 commonly grounded elements.

3.6.5 Delete 3.6.5 in its entirety and substitute the following:

3.6.5 Detectivity. The 500K blackbody detectivity of all nondefective elements shall be greater than 8.5×10^{9} cm $Hz^{\frac{1}{2}}$ W^{-1}. The average detectivity of all elements of the array which are not hard defects shall be greater than 1.1×10^{10} cm $Hz^{\frac{1}{2}}$ W^{-1}.

3.6.5.1, Line 2. Delete "3.5" and substitute "2.5".

3.6.8, Line 5. Delete "1 mA ±10 percent" and substitute "1.00 mA, ±0.01 mA."

3.6.9, Line 2. Delete "7.7, ±0.25 to 11.75, ±0.25 micrometers" and substitute "7.7, ±0.3 to 11.7, ±0.5 micrometers."

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4.6.9 Delete 4.6.9 in its entirety and substitute the following:

4.6.9 Spectral response. The spectral response of element numbers 36, 91, and 146 shall be tested per "Standard Procedures for Testing Infrared Detectors and Describing Their Performance," by R. C. Jones, D. Goodwin, and G. Pullen, dated 12 September 1960, Office of the Director of Defense Research and Engineering. If these are hard defects, an adjacent element may be substituted. The spectral sensitivity is defined as the 50 percent relative response points and shall be measured on the assembled detector-Dewar. Failure to meet requirements of 3.6.9 shall constitute failure of this test.

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4.6.14 Delete 4.6.14 in its entirety and substitute the following:

4.6.14 Microphonics. The detector-Dewar shall be coupled to a mechanical cryogenic cooler (MIL-C-49175) or equivalent and operated in the cooled condition of 60-80 K with the channels under test at full bias. The vibrational excitation shall be turned off to establish the white noise level. All elements shall be checked using a noise power spectrum analyzer with 50, ±10 Hz noise bandwidth between 50 Hz and 20 kHz. Any element which fails to meet the requirements of 3.6.14 shall be considered a hard defect.

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6.2 Add line k as follows:

"k. MIL-STD-781B shall be used for reliability testing of 4.8."

6.3.2 Delete 6.3.2 in its entirety and substitute the following:

6.3.2 Defective element. A defective element shall be defined as an element which fails to meet any of the requirements of 3.6.5, 3.6.6, 3.6.7, and 3.6.14. A defective element may be

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classified as either hard or soft. A hard defect is an element which fails the requirements of 3.6.6, 3.6.7 or 3.6.14 or which has a 500K blackbody detectivity less than 5.0×10^9 cm $Hz^{\frac{1}{2}} W^{-1}$. Two elements shorted together are considered one hard defect. A soft defect is one which meets all requirements of 3.6.6, 3.6.7 and 3.6.14 and has a 500K blackbody detectivity greater than 5.0×10^9 cm $Hz^{\frac{1}{2}} W^{-1}$ but less than 8.5×10^9 cm $Hz^{\frac{1}{2}} W^{-1}$.

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